

Individual Risk Estimation in μ -Argus: A Review

Luisa Franconi and Silvia Polettini

ISTAT

Servizio Progettazione e Supporto Metodologico
nei Processi di Produzione Statistica
Via Cesare Balbo 16, 00184 Roma, Italy
{franconi,polettin}@istat.it

Abstract. Individual risk estimation was one of the issues that the European Union project CASC targeted. On this subject ISTAT has built on previous work by Benedetti and Franconi (1998) to improve individual risk measures. These permit the identification of unsafe records to be protected by disclosure limitation techniques. The software μ -Argus contains now a routine, that has been implemented by CBS Netherlands in cooperation with ISTAT, for computing the Benedetti-Franconi or individual risk of disclosure. The paper reviews the main aspects of the individual risk methodology. Such an approach defines measures of risk for protection of files of independent records as well as hierarchical files. The theory and some practical issues such as threshold setting are illustrated for both cases.

Keywords: Argus software, data protection, global risk, individual risk, household data, local suppression, re-identification disclosure, threshold setting, unsafe combinations

1 Introduction

One of the main objectives of the European Union project IST-2000-25069 CASC on Computational Aspects of Statistical Confidentiality was to develop a free software for data protection. For files of microdata, μ -Argus see [Hun02] aims to support users in charge of data release during the phase of data protection. The software builds further on the version released under the Statistical Disclosure Control (SDC) project within the European Union's 4th Framework Program. Among the features of μ -Argus, the user can apply *local suppression* to selected records or *global recoding* to the whole file see [WdW01]. Measures of risk per record can be exploited in the process of data protection to judge where to use local suppression or to assess the effectiveness of the global recoding; for this reason per-record risk estimation was one of the issues that the CASC project targeted. On this subject ISTAT has built on previous work by [BF98] to improve one such risk measure, that is used to identify unsafe records. The software μ -Argus contains now a routine, that has been implemented by CBS

Netherlands in cooperation with ISTAT, for computing the *Benedetti-Franconi* (B-F) or *individual risk* of re-identification.

This paper reviews the main aspects of the B-F individual risk methodology. We discuss measures of risk for files of independent records as well as for *hierarchical files*. We limit our discussion to methodological aspects of the procedure, while referring to [PS03] for the practice of the individual risk methodology within μ -Argus, and to the Argus manual for an overview of the software¹.

A relevant characteristic of social microdata is its inherent hierarchical structure, which allows us to recognise groups of individuals in the file, the most typical case being the household. When defining the re-identification risk, it is important to take into account this dependence among units: indeed re-identification of an individual in the group may affect the probability of disclosure of all its members.

So far, implementation of a hierarchical risk has been performed only with reference to households. We will therefore refer to individual and household risk for files of independent units and hierarchical files, respectively.

The individual risk methodology is currently the only approach that allows for a hierarchical structure in the data, which is in fact typical in many contexts. Allowing for dependence in estimating the risk enables us to attain a higher level of safety than when merely considering the case of independence.

The paper is organised as follows: in Sect. 2 we discuss measures of risk of re-identification. In particular, in Sect. 2.2 and 2.3 we define the individual risk and discuss its estimation using sample data, whereas in Sect. 2.4 we introduce the household risk, use of which is appropriate when a hierarchical structure is present in the microdata. Section 3 is devoted to the combined use of these measures of risk and local suppression to protect a file as it is currently implemented in μ -Argus. In order to identify unsafe records, a risk threshold is to be selected. Section 3.1 describes the tools that μ -Argus offers to perform this task, whereas Sect. 3.2 presents methods for threshold setting using global risk measures. Finally, in Sect. 4 we present our conclusions.

2 Extending the Measure of Risk from Individual to Household Samples

In this section we discuss two related measures of disclosure risk, both implemented in μ -Argus. The first (the individual risk) applies to files of independent records, whereas the second (the household risk) suits hierarchically structured, e.g. household, microdata. Under this circumstance it is possible to locate units within the household and, to a certain extent, also establish relationships between members of the same household. The household risk is defined in terms of the individual risk; the rationale behind such a definition is that if a household is re-identified, its members can be re-identified.

The definition of our risk measures is based on the concept of re-identification disclosure e.g. [CKM98,FM98,SH98,DL86,WdW01], and is appropriate for sam-

¹ Available at <http://neon.vb.cbs.nl/CASC/MU.html>

ples of microdata stemming from social surveys. By record re-identification we mean that the unit in the released file and a unit in the register that an intruder has access to belong to the same individual in the population. The underlying hypothesis is that the intruder will always try to match a record and a unit in the register (i.e. if the record is not unique he will try a probabilistic link), and that he will use public domain variables (*key variables*) only. As external registers that contain information on households are not commonly available, for files of households we make the assumption that the intruder attempts a confidentiality breach by re-identification of individuals.

In social survey data, categorical key variables allow us to tackle the problem of disclosure limitation via the concept of unique or rare combinations in the sample. A key issue is to be able to distinguish between combinations that are at risk, for example sample uniques corresponding to rare combinations in the population, and combinations that are not at risk, for example sample uniques corresponding to combinations that are common in the population. To this aim, a step of inference from the sample to the population is performed. Instead of focusing on the *sample* frequency of combinations of key variables, the individual risk of disclosure can be defined as the probability that a sampled record is re-identified (i.e. recognised as corresponding to a particular unit in the population). This value can then be estimated for each record in the released file on the basis of the observed sample. This approach would allow for a more parsimonious use of suppressions than the traditional local suppression methodology based on the sample frequency of combinations. The latter is present in μ -Argus since the European Union project SDC see [WdW01]. See [FP03] for a comparison.

In the last few years a number of proposals have been made: [FM98], [SH98] and [ES04] define, with different motivations, a log linear model for the estimation of the individual risk. [BF98] propose a methodology to estimate a measure of risk per record using the sampling weights, as the usual instrument that National Statistical Institutes adopt to allow for inference from the sample to the population. Further discussion of the approach is in [DCFS03], [Pol03], [Rin03]. A related approach is described in [Car02].

2.1 Some Notation

Let the released file be a random sample s of size n drawn from a finite population P consisting of N units. For a generic unit i in the population, we denote w_i its final sampling design weight. Under the hypothesis that the key variables are discrete, cross-tabulating these produces a set of combinations $\{1, \dots, K\}$. A combination k is defined to be the k -th cell in the cross-tabulation. The set of combinations defines a partition of both the population and the sample, and the values of the key variables observed on unit i will classify such a record into one combination. We denote by $k = k(i)$ the cell into which the sampled record i falls. Let f_k and F_k denote the size of the k -th cell in the sample and population, respectively. Retaining only the observed combinations -combinations with zero sample frequency being omitted- does not alter the above partition of the sample.

2.2 Definition of the Individual Risk

Assume for simplicity that there is complete agreement between the sample and the external archive available to the intruder, as far as the key variables are concerned (for a more general setting, see [Pol03]. We first note that if we were to know the population frequency of the k -th combination, F_k , we would define the probability of re-identification simply by $1/F_k$, for each record that is classified in combination k (i.e. $\forall i : k(i) = k$). The population frequencies are generally unknown, therefore an inferential step is to be performed. In the proposal by [BF98] the uncertainty on F_k is accounted for in a Bayesian fashion by introducing the distribution of the population frequencies given the sample frequencies. The individual risk of disclosure is then measured as the (posterior) mean of $1/F_k$ with respect to the distribution of $F_k|f_k$:

$$r_k = E\left(\frac{1}{F_k} | f_k\right) = \sum_{h \geq f_k} \frac{1}{h} \Pr(F_k = h | f_k) \quad . \quad (1)$$

To determine the probability mass function of $F_k|f_k$, the following superpopulation approach is introduced [BKP90,Rin03,Pol03]:

$$\begin{aligned} \pi_k &\sim [\pi_k] \propto 1/\pi_k, \quad \pi_k > 0, & k=1, \dots, K, & \text{independently} \\ F_k | \pi_k &\sim \text{Poisson}(N\pi_k), & F_k=0, 1, \dots, & \text{independently} \\ f_k | F_k, \pi_k, p_k &\sim \text{binomial}(F_k, p_k), & f_k=0, 1, \dots, F_k, & \text{independently} \end{aligned} \quad (2)$$

Under these assumptions, the posterior distribution of $F_k|f_k$ is negative binomial with success probability p_k and number of successes f_k . In general, the probability mass function of a negative binomial variable counting the number of trials before the j -th success, each with probability p_k , is the following:

$$\Pr(F_k = h | f_k = j) = \binom{h-1}{j-1} p_k^j (1-p_k)^{h-j}, \quad h \geq j \quad .$$

In [BF98] it is shown that under the negative binomial distribution the risk (1) can be expressed as

$$r_k = E(F_k^{-1} | f_k) = \int_0^\infty \left\{ \frac{p_k \exp(-t)}{1 - q_k \exp(-t)} \right\}^{f_k} dt \quad .$$

The transformation $\exp(-t) = y$ gives the integral

$$r_k = p_k^{f_k} \int_0^1 t^{f_k-1} (1 - tq_k)^{-f_k} dt$$

that can be expressed [Pol03] via the Hypergeometric function ${}_2F_1(a, b; c; z)$ as

$$r_k = \frac{p_k^{f_k}}{f_k} {}_2F_1(f_k, f_k; f_k + 1; q_k) \quad (3)$$

where

$$F(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 t^{b-1} (1-t)^{c-b-1} (1-tz)^{-a} dt$$

is the integral representation (valid for $\Re(c) > \Re(b) > 0$) of the Gauss Hypergeometric series [AS65].

Substitution of an estimate of p_k in (3) leads to an estimate of the individual risk of disclosure (1). Given F_k , the maximum likelihood estimator of p_k under the binomial model in (2) is $\hat{p}_k = f_k/F_k$. F_k being not observable, [BF98] propose to use

$$\hat{p}_k = \frac{f_k}{\sum_{i:k(i)=k} w_i},$$

where $\sum_{i:k(i)=k} w_i$ is an estimate of F_k based on the sampling design, possibly calibrated [DS92].

The negative binomial distribution is defined for $0 < p_k < 1$; in practice, estimates $\hat{p}_k$ might attain the extremes of the unit interval. We never deal with $\hat{p}_k = 0$, as it corresponds to $f_k = 0$. On the other hand, when $\hat{p}_k = 1$, ${}_2F_1(f_k, f_k, f_k + 1, 0) = 1$, therefore the individual risk equals $1/f_k$.

2.3 Some Comments on Risk Estimation

In [DCFS03] an experiment was conducted to assess the performance of the individual risk of disclosure. The aim was to investigate whether the individual risk is estimating the correct quantity, i.e. the real risk of an individual, and also whether the quality of this estimation is appropriate. A good agreement was noticed between the real risk and its estimates, although the precision of the estimator seems poor for rare combinations in the sample as compared to the more common ones. A minor precision is an intrinsic problem of small counts. Discriminating rare and common features in the population is, by far, the most difficult task especially when one can count on only one occurrence in the sample. For this reason further studies to improve the performance of the estimator used for the individual risk have been planned.

Model (2) was questioned to provide a good fit to real data under some circumstances [Rin03]. Our experiments seem to indicate that the model holds, at least with our data. Notice that the assumed model is compatible with a large number of relatively small cells in the population. These might occur with both small sized population and large number of combinations of key variables.

2.4 Household Risk

From the definition given in Sect. 2.2, it follows that the individual risk can be considered an estimate of a probability of re-identification. We define the household risk as the probability that at least one individual in the household is re-identified. By consequence, the household risk can be derived from the individual risks and knowledge of the household structure of the data.

For a given household g of size $|g|$, whose members we label $i_1, \dots, i_{|g|}$, the *household risk* is defined as

$$r_g^h = Pr(i_1 \cup i_2 \cup \dots \cup i_{|g|} \text{ re-identified}) ; \quad (4)$$

Assuming independence of re-identification attempts within the same household, (4) can be expressed by Boole's formula using the individual risks $r_{i_1}, \dots, r_{i_{|g|}}$ defined in (1):

$$r_g^h = \sum_{j=1}^{|g|} r_{i_j} - \sum_{i_j < i_l} r_{i_j} r_{i_l} + \sum_{i_j < i_l < i_m} r_{i_j} r_{i_l} r_{i_m} + \dots + (-1)^{|g|+1} r_{i_1} r_{i_2} \dots r_{i_{|g|}} . \quad (5)$$

By symmetry, the ordering of units in the group is not relevant. In a hierarchical file therefore the measure of disclosure risk is the same for all the individuals in household g and equals r_g^h .

3 Protection

The *risk model* implemented in μ -Argus permits to classify records into safe or unsafe according to their individual risk and, if relevant, the household structure of the file. μ -Argus then applies local suppression [dWW98]see to the unsafe records only, or global recoding to the whole file. To apply local suppression users must select a threshold, e.g. a level of acceptable risk, representing a risk value below which an individual can be considered safe. Global recoding can be assessed by the same threshold; such a procedure can be repeated until all records in the file are safe.

3.1 Direct Specification of Risk Thresholds

Having selected the set of key variables and the individual risk methodology, μ -Argus returns a graphical facility showing the histogram of the individual risk (see the top panel of Fig. 1). For data organised according to a household structure, a similar graph will be produced for the household risk (bottom panel of Fig. 1). For files of independent records, a threshold on the individual risk can be selected either manually, entering a value in the **Risk threshold** box, or through the slider at the bottom of the risk histogram. In both panels of Fig. 1, the **Calc** button allows the user to compute the number of unsafe records (households) corresponding to the given threshold. Conversely, the user can specify the number of unsafe records (households) and determine the corresponding risk threshold via the corresponding **Calc** button.

But for providing the number of unsafe records (households), so far μ -Argus provides no guidance on how to select a risk threshold; a criterion is discussed in Sect. 3.2, first proposed in [PS03] and [Pol03] for non-hierarchical files. Throughout this section we assume for simplicity that the user has in mind a target threshold level, possibly chosen by the means to be described in Sect. 3.2.

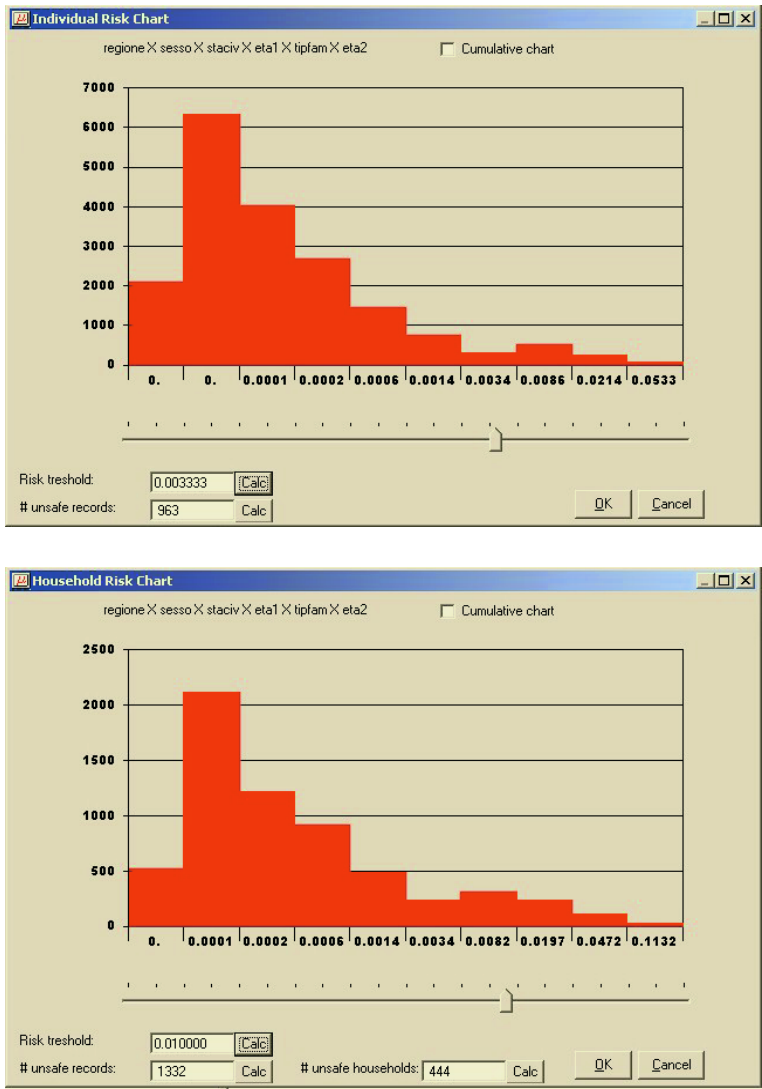


Fig. 1. Risk histograms: individual (top panel) and household risk (bottom panel). A logarithmic scale is used

For files of independent records, setting a threshold on the individual risk is a straightforward operation that can be accomplished as discussed previously. Next we discuss how to select thresholds when dealing with household data. We therefore assume that the user has in mind a threshold α on the *household risk* r_g^h . We first notice that each record belonging to household g has the same disclosure risk r_g^h . By definition, the hierarchical risk of household g , r_g^h , is bounded by the sum of the individual risks of records belonging to household g (see (5)):

$$r_g^h \leq \sum_{j=1}^{|g|} r_{i_j} \quad .$$

For a threshold α , household g is safe if $r_g^h < \alpha$; in order for household g to be classified safe, it is therefore sufficient that each of its components has individual risk less than $\delta_g = \alpha/|g|$. Note that δ_g varies across individuals according to the household they belong to. In practice, denoting by $g(i)$ the household to which record i belongs, a threshold α on the household risk is turned into a vector of thresholds on the *individual risks* r_i , $i = 1, \dots, n$:

$$\delta_g = \delta_{g(i)} = \alpha/|g(i)| \quad . \quad (6)$$

Clearly this is a strongly prudential approach. Records are therefore set to unsafe whenever $r_i > \delta_{g(i)}$; local suppression is then applied to those records, if requested. Suppression ensures that after protection the household risk is below the threshold α .

This approach to threshold setting for identifying unsafe records in hierarchical files has been implemented in the latest beta-release of μ -Argus.

3.2 Methods for Threshold Setting via Global Risk Measures

As discussed in the previous section, threshold setting results in a (possibly household-specific) threshold level of the individual risk. Argus provides no guidance to the user on how to perform this task; [PS03] and [Pol03] propose to use the so-called *re-identification rate*.

The individual risk provides a measure of *risk per record*. A *global* measure of disclosure risk for the whole sample can be expressed by the re-identification rate, which is defined by means of the expected number of re-identifications in the file. Global measures of this kind are not new in the literature of disclosure limitation, see for example [Lam93] and [DFT02]. Here this measure is discussed as a tool for setting a risk threshold and identify unsafe records.

We refer to the interpretation of r_k as an estimate of a probability of re-identification to introduce the expected number of re-identifications in the file,

$\sum_{k=1}^K r_k f_k$; the re-identification rate is then

$$R = \frac{1}{n} \sum_{k=1}^K r_k f_k \quad .$$

One use of global risk measures is the following: if the re-identification rate of the file is below a level η that the user considers acceptable, i.e. $R < \eta$, then the file can be released. As R depends on r_k s, a threshold α can be determined for the individual risk, such that R does not exceed η . From a user's perspective, working on rates or percentages of re-identifications in the file is easier and intuitively more appealing than working on the risk scale. Moreover use of a

global measure such as R also implies an evaluation on the overall safety of the released file.

A similar reasoning applies when setting the threshold for the household risk. Denote by G the total number of households in the file, and define the expected number of re-identified *households* in the file as $\sum_{g=1}^G r_g^h$. The household re-identification rate

$$R^h = \sum_{g=1}^G r_g^h / G$$

can be exploited to select a household risk threshold, along the same lines as above. Once a threshold α on the household risk has been selected, the procedure described in Sect. 3.1 can be applied to transform that value into an individual risk threshold (see (6)).

Threshold setting via the re-identification rate is currently in the process of being implemented into Argus.

4 Conclusions

Individual risk estimation was one of the issues that the European Union project IST-2000-25069 CASC on Computational Aspects of Statistical Confidentiality targeted. On this subject ISTAT has built on previous work by [BF98] to derive individual risk measures that can be exploited in the process of data protection to identify unsafe records. One of the main project's results is the software Argus, a practical tool aimed to support users in charge of data release during the phase of data protection. In particular the software μ -Argus, specific to micro-data, contains now a routine, that has been implemented by CBS Netherlands in cooperation with ISTAT, for computing the *Benedetti-Franconi* (B-F) or *individual risk* of disclosure. We have reviewed in the paper the main results that the project has achieved as far as the individual risk methodology is concerned.

Implementation of the individual risk methodology into the free software μ -Argus has made its application easier and allowed testing of the procedure and comparisons with other methods. During Argus' testing phase researchers not directly involved in the CASC project have been exposed to theory and application of these techniques, some of whom were new in the literature. In particular, a debate on risk models has emerged, concerning the underlying assumptions of the individual risk model and the assessment of the theory itself; alternative measures of risk and approaches to estimation have been proposed or called for [Rin03]. Such a debate had a beneficial effect in suggesting refinements of the theory and directions for future work; the theory on the individual risk has been further examined [DCFS03, Pol03], and extensions and modifications are currently under study by the authors; other approaches to risk estimation are being pursued e.g. by [Rin03] (where measures of performance are also mentioned), [Car02], [ES04].

Further research to improve the performance of the estimator used for the individual risk has been planned; in particular, it is foreseen that the use of

methodologies borrowed from the area of small domain estimation will allow for progress in this field.

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